

INGUVA SATYAVENAKATA SAI KUMAR

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Experience

NStarX

Jun 2024 – Present

Software Engineer

- Built GenAI Lab as a Micro Frontend in React.js where each AI service (RAG, Multimodal, and Report Generator) has its own frontend and Python backend with independent APIs and deployments, all under one unified platform for SHI.
- Built real-time pod and cluster monitoring dashboards in Vue.js for Digital Human, showing live metrics, resource usage, and logs so the team could always see what was running, what was consuming resources, and where things failed.
- Used AWS S3 for file storage in the HRMS, and worked with EC2 and Lambda on an ongoing project to handle compute provisioning and event-based processing in the cloud, supporting scalable and reliable service deployments in production.
- Wrote tests using pytest and unittest for Python backend services and Jest for React frontend components to catch bugs early, avoid regressions, and keep the codebase stable across the full stack as the project continued to grow.
- Built a full-stack HRMS in Python and Frappe Framework serving 100+ users, working in a cross-functional team of 2 and delivering all core HR modules on time using Git-based version control and Agile sprint workflows throughout.

Projects

GenAI Lab

Jan 2026 – Present

- Architected GenAI Lab, a microservices-based Micro Frontend control plane in React.js for SHI — decomposing RAG, Multimodal, and Report Generator into independently deployable services with isolated ports and role-based access.
- Engineered the Admin-User access control layer where admins provision GenAI microservices per user role, triggering one-click ArgoCD deployments on Kubernetes that spin up isolated, containerized services instantly via Docker.
- Developed a dual-mode versioning flow in GenAI Lab — users instantly try a live version or deploy a custom one — validated through pytest unit tests and accelerated using AI coding assistants throughout the development lifecycle.

Tech Stack: React, AWS, GitLab CI/CD, Python

Film Sense

Feb 2026 – Present

- Built FilmSense, a full stack movie discovery app enabling semantic search by Genre, Character, and Theme using Qdrant vector database and Sentence Transformers embeddings, with fuzzy matching and guardrails for accurate results.
- Developed a Movie Details page aggregating overview, director, writers, cast, ratings, votes, runtime, and trailer via FastAPI backend and React/TypeScript frontend, plus a live dashboard monitoring CPU, RAM, disk, and GPU usage.

Tech Stack: React, FastAPI, Python, TypeScript, Hugging Face Transformers

Digital Human

Sep 2025 – Jan 2026

- Developed a real-time metrics dashboard in Vue.js for the Digital Human client platform — combining polling and event streams to track live CPU, memory, and pod health, with popup log viewers per pod and cluster for instant debugging.
- Engineered an iframe-based service deployment layer within Digital Human — embedding and launching each AI service directly inside the platform without external redirects, enabling seamless in-app service access for end users.

Tech Stack: Vue.js, Python, Docker, Kubernetes

Skills

- **Languages:** JavaScript, TypeScript, Python, HTML, CSS, Node.js
- **Frameworks:** React, FastAPI, Flask, Next.js, Vue.js, System Design, RestAPI
- **Tools:** Docker, AWS Redshift, GitHub, AWS S3 Bucket, AWS Lambda, AWS CodeCommit, GitLab CI/CD
- **Databases & AI Tools:** RAG, MultiModal, PostgreSQL, MySQL
- **Testing Tools:** pytest, Jest, Vitest

Education

Gitam University

Oct 2020 – Apr 2024

B.Tech in Electronics and Communication Engineering: CGPA: 8.10

Hyderabad